

Module 13 - Seed Data, Testing & Deployment

Make the app deterministic, demo-ready, and safe to ship

SOURCE OF TRUTH: This resource intentionally uses plain JavaScript. The locked stack is React + JavaScript + Firebase + Material UI (MUI). Do not introduce Next.js, TypeScript, shadcn/ui, Redux, or another backend framework unless the project owner explicitly changes this specification.

Scope

Provide realistic demo data, test critical business logic, and define the deployment checklist. The app should feel populated for demos without weakening production security.

Seed data

- 3-5 demo users
- 3 demo groups
- 4-6 plans across different types
- Several bookings
- A realistic expense graph with balances
- At least one poll with votes
- Activity feed entries
- A mix of completed/upcoming/cancelled records

Demo scenarios

Scenario A: Movie Night

- 5 members
- movie booking
- equal split

Scenario B: Goa Trip

- 6 members
- hotel + flight + activities
- multiple payers
- non-trivial settlement

Scenario C: Restaurant Decision

- poll
- realtime voting
- selected option
- expense after dinner

Testing priorities

- Authentication route protection
- Firestore rule authorization
- Expense split calculations
- Settlement algorithm
- Invite expiry/revocation
- Poll duplicate voting
- Responsive layout smoke tests
- Optimistic update rollback

Do not fake critical flows

A demo can contain realistic seeded records, but buttons that claim to save, delete, settle, join, vote, or update data must perform the real Firebase operation. Avoid hard-coded “success” UI for core actions.

Deployment checklist

- Production Firebase project selected
- Auth providers configured
- Firestore rules deployed
- Required Firestore indexes deployed
- Storage rules deployed
- Environment configuration separated from source
- Demo seed not automatically executed in production
- Error logging enabled
- Build tested on desktop + mobile
- Security rules tested before launch

Acceptance criteria

- Fresh deployment loads without manual database edits.
- Demo account/scenario is repeatable.
- Core business logic tests pass.
- Production rules prevent cross-group access.
- No secrets are committed to source control.